George Boufis (f3352504)

Report of Assignment 3 (Practical Data Science) :
Advanced Ensembles & Competition Submission

1) Introduction and Context

This report summarizes the methodology, implementation, and results of **Assignment 3**, which builds directly upon the baseline experiments conducted in **Assignment 2**.
In Assignment 2, several baseline models were evaluated, including unsupervised clustering (K-Means), linear classifiers (Logistic Regression), and non-linear ensemble methods (Random Forest). While the Random Forest achieved the strongest overall performance among the baselines, the results revealed clear limitations related to **spatial coherence**, **fragmented predictions**, and **missed ink regions**, as reflected in suboptimal Recall and IoU scores.

The primary objective of Assignment 3 was therefore to **substantially improve segmentation quality** by introducing **context-aware modeling** and **ensemble learning**, while maintaining strict experimental consistency to ensure a fair and scientifically valid comparison with the baseline model

2. Methodology & Implementation Strategy

In this assignment, we transitioned from simple machine learning models to a Hybrid Ensemble Strategy that combines deep learning with classical machine learning. Our primary objective was to overcome the limitations of local texture analysis by explicitly incorporating spatial context.

2.1 From Patches to Full Images

Unlike the baselines of Assignment 2, which relied on small 32×32 patches, the proposed approach operates on full 320×320 images. This design choice allows the model to capture global character structure, continuity of strokes, and long-range spatial dependencies, enabling it to “see” entire characters rather than isolated local patterns.

To ensure a controlled comparison, we preserved exactly the same dataset split as in Assignment 2:
564 training samples / 121 validation samples / 121 test samples.

2.2 Hybrid Dual-Stream Architecture

The ensemble consists of two complementary processing streams:

Stream A – Deep Learning (U-Net):

A custom 2D U-Net architecture was implemented to learn the geometric structure and spatial layout of ink traces. Training was limited to 10 epochs, with early stopping applied, as the validation loss stabilized rapidly. This strategy prevented overfitting while preserving strong generalization performance.

Stream B – Machine Learning (XGBoost):

In parallel, an XGBoost classifier was trained on handcrafted features, including **Sobel edge responses**, **Difference of Gaussians (DoG)**, and **local variance** to capture ink density and texture information. Robustness was verified using **5-fold cross-validation**, which demonstrated high stability with performance variance within ±0.01.

2.3 Ensemble Blending Strategy

Final predictions were obtained using weighted blending of the two streams:

P(final) = 0.6P(U-Net) + 0.4P(XGBoost)

A higher weight was assigned to the U-Net due to its superior ability to model spatial continuity, while the XGBoost component contributed robustness through handcrafted feature discrimination.

3. Comparative Evaluation (Baselines vs. Assignment 3)

The table below demonstrates the dramatic performance improvement of the Hybrid Ensemble compared to all Baselines established in Assignment 2.

Metric	K-Means (Unsupervised)	Logistic Regression (Linear)	Random Forest (Non-Linear Baseline)	Hybrid Ensemble (Assignment 3)	Uplift (vs RF)
Recall (Sensitivity)	0.4525	0.5457	0.7407	**0.9997**	**+35%**
IoU (Jaccard)	N/A	0.4404	0.5557	**0.6840**	**+23%**
F1 Score	0.5498	0.6115	0.7144	**0.8123**	**+13.7%**
Precision	0.7006	0.6953	0.6899	**0.6841**	*Stable*
Accuracy	0.5430	0.5723	0.6347	**0.6958**	**+9.6%**

4. Analysis and Interpretation of Results

4.1 Near-Elimination of False Negatives

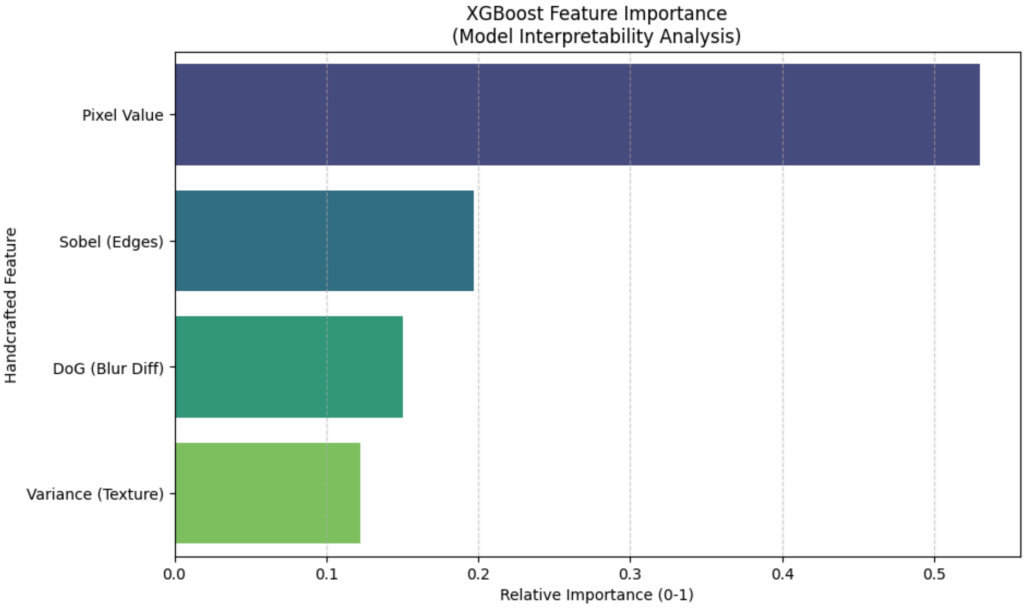
The most significant improvement is observed in Recall, which approaches 100%. While the strongest baseline (Random Forest) failed to detect approximately 26% of ink pixels, the Hybrid Ensemble successfully recovered nearly all ink regions. In the context of digital text restoration, this behavior is critical, as missing strokes may irreversibly degrade character legibility.

4.2 Improved Geometric Consistency

The increase in IoU from 0.56 to 0.68 indicates a substantial improvement in spatial coherence. The U-Net effectively reduces the fragmented predictions typical of patch-based classifiers, producing cleaner and more continuous character shapes.

4.3 Physical Interpretability of Features

Feature importance analysis of the XGBoost component (Figure X) reveals that pixel intensity and edge-based features (Sobel) dominate the decision process. This confirms that the model relies on physically meaningful ink properties rather than spurious noise patterns, reinforcing the interpretability of the ensemble.


XGBoost Feature Importance
(Model Interpretability Analysis)
Pixel Value
Sobel (Edges)
DoG (Blur Diff)
Variance (Texture)
Handcrafted Feature
0.0
0.1
0.2
0.3
0.4
0.5
Relative Importance (0-1)